

ANALOG CMOS CIRCUIT DESIGN SS26

PRACTICAL EXERCISE I

T3: Transistor Transfer Characteristic Curve

- a) Simulate the I_D versus U_{DS} characteristic curve of an nmos-Transistor ("nmos" from library "PRIMLIB") for $0V < V_{DS} < 3.3V$, $V_{GS} = 0.5V, 0.6V, \dots 1.2V$, a transistor length L equal to $0.35\mu m$ and $1\mu m$ and a transistor width W equal to $2\mu m$. Attach the simulation (parametric sweep) results to the documentation.
- b) Derive an equation which allows for the determination of the channel length modulation parameter λ (lambda) from the simulation results.
- c) Determine λ for the parameter sets of V_{GS} and L given in (a). What can be observed?

